

List of PWS NASA Dataset files:

PWS NASA Dataset	Dataset Description	List of files
111	Case 1 Simulation Time 11 Wet Microburst with Rain and Hail, 08/02/85 DFW Accident Case, Dallas-Fort Worth, Texas	PWS_NASA_Dataset111_hail.faa PWS_NASA_Dataset111_rain.faa PWS_NASA_Dataset111_rrf.faa PWS_NASA_Dataset111_tau.faa PWS_NASA_Dataset111_u.faa PWS_NASA_Dataset111_v.faa PWS_NASA_Dataset111_w.faa PWS_NASA_Dataset111_xiv.faa
237	Case 2 Simulation Time 37 Wet Microburst, 06/20/91 NASA Research Flight, Orlando, Florida	PWS_NASA_Dataset237_rain.faa PWS_NASA_Dataset237_rrf.faa PWS_NASA_Dataset237_tau.faa PWS_NASA_Dataset237_u.faa PWS_NASA_Dataset237_v.faa PWS_NASA_Dataset237_w.faa PWS_NASA_Dataset237_xiv.faa
349	Case 3 Simulation Time 49 Multiple Microburst, 07/11/88 Incident Case, Denver, Colorado	PWS_NASA_Dataset349_hail.faa PWS_NASA_Dataset349_rain.faa PWS_NASA_Dataset349_rrf.faa PWS_NASA_Dataset349_tau.faa PWS_NASA_Dataset349_u.faa PWS_NASA_Dataset349_v.faa PWS_NASA_Dataset349_w.faa PWS_NASA_Dataset349_xiv.faa
351	Case 3 Simulation Time 51 Multiple Microburst, 07/11/88 Incident Case, Denver, Colorado	PWS_NASA_Dataset351_hail.faa PWS_NASA_Dataset351_rain.faa PWS_NASA_Dataset351_rrf.faa PWS_NASA_Dataset351_tau.faa PWS_NASA_Dataset351_u.faa PWS_NASA_Dataset351_v.faa PWS_NASA_Dataset351_w.faa PWS_NASA_Dataset351_xiv.faa
436	Case 4 Simulation Time 36 Warm Microburst, 07/14/82 Sounding Stable Layer, Denver, Colorado	PWS_NASA_Dataset436_rain.faa PWS_NASA_Dataset436_rrf.faa PWS_NASA_Dataset436_tau.faa PWS_NASA_Dataset436_u.faa PWS_NASA_Dataset436_v.faa PWS_NASA_Dataset436_w.faa PWS_NASA_Dataset436_xiv.faa
540	Case 5 Simulation Time 40 Very Dry Microburst, 07/08/89 Sounding, Denver, Colorado	PWS_NASA_Dataset540_rain.faa PWS_NASA_Dataset540_rrf.faa PWS_NASA_Dataset540_rrfi.faa PWS_NASA_Dataset540_tau.faa PWS_NASA_Dataset540_u.faa PWS_NASA_Dataset540_v.faa PWS_NASA_Dataset540_w.faa PWS_NASA_Dataset540_xiv.faa

PWS NASA Dataset	Dataset Description	List of files
545	Case 5 Simulation Time 45 Extremely Dry Microburst (Second Pulse), 07/08/89 Sounding, Denver, Colorado	PWS_NASA_Dataset545_rain.faa PWS_NASA_Dataset545_rrf.faa PWS_NASA_Dataset545_rrfi.faa PWS_NASA_Dataset545_tau.faa PWS_NASA_Dataset545_u.faa PWS_NASA_Dataset545_v.faa PWS_NASA_Dataset545_w.faa PWS_NASA_Dataset545_xiv.faa
614	Case 6 Simulation Time 14 Highly Asymmetric Microburst, Derived Sounding, Florida	PWS_NASA_Dataset614_hail.faa PWS_NASA_Dataset614_rain.faa PWS_NASA_Dataset614_rrf.faa PWS_NASA_Dataset614_rrfi.faa PWS_NASA_Dataset614_tau.faa PWS_NASA_Dataset614_u.faa PWS_NASA_Dataset614_v.faa PWS_NASA_Dataset614_w.faa PWS_NASA_Dataset614_xiv.faa
727	Case 7 Simulation Time 27 Gust Front, 08/02/81 Adjusted Knowlton Sounding, Montana	PWS_NASA_Dataset727_cld.faa PWS_NASA_Dataset727_hail.faa PWS_NASA_Dataset727_rain.faa PWS_NASA_Dataset727_rrf.faa PWS_NASA_Dataset727_rrfi.faa PWS_NASA_Dataset727_tau.faa PWS_NASA_Dataset727_u.faa PWS_NASA_Dataset727_v.faa PWS_NASA_Dataset727_w.faa PWS_NASA_Dataset727_xiv.faa

Variables needed for PWS testing:

Note: Either RRF or RRFI (if available) can be used as the radar reflectivity factor of precipitation.

Variable	Description	Units
U	Horizontal component of wind velocity directed east and west (Positive values in the increasing x direction)	meters per second
V	Horizontal component of wind velocity directed north and south (Positive values in the increasing y direction)	meters per second
W	Vertical component of wind velocity (Positive values directed upwards)	meters per second
RRF	Radar reflectivity factor of precipitation	decibels of Z (dBZ)
RRFI	Radar reflectivity factor of precipitation including insects <i>Only included in datasets 540, 545, 614, and 727</i>	decibels of Z (dBZ)

Additional variables:

Variable	Description	Units
TAU	Temperature	degrees Kelvin
XIV	Water vapor	grams per cubic meter
RAIN	Rain water content	grams per cubic meter
HAIL	Hail water content <i>Only included in datasets 111, 349, 351, 614, and 727</i>	grams per cubic meter
CLD	Liquid cloud-droplet water <i>Only included in dataset 727</i>	grams per cubic meter

Information

- The data set grid is a right-handed x, y, z Cartesian coordinate system with x pointing east, y pointing north, and z directed upwards.
- The velocity components, U, V, and W correspond to the x, y, and z directions, respectively.
- Each variable, located in a separate file, is a three-dimensional spatial array of data.

Data contained in the five header lines of each file:

Variable	Description
title information	general information about the dataset
variable name	data in the file is for this variable
ix	number of grid points in x
iy	number of grid points in y
iz	number of grid points in z
simtime	time of the simulation in seconds
xstart	x origin in meters
ystart	y origin in meters
zstart	z origin in meters
dx	grid cell size for x in meters
dy	grid cell size for y in meters
dz	grid cell size for z in meters

How to read the data field format

Each ASCII data file begins with a five-line header to describe the size, the time, the location, and the resolution of the three-dimensional domain. For variables describing a three-dimensional quantity (position, size, resolution), the order of the coordinates are x, y, and then z.

The first line contains a descriptive title for the data set. The next line lists which variable this file contains. The third line contains the integers defining the size of the three-dimensional domain. The fourth line starts with the time of the simulation in seconds followed by the origin of the three-dimensional domain. The final line before the data field is the grid resolution. The three-dimensional data with six numbers per line then follow these header lines. The following example in FORTRAN shows how to read a single data file.

```
      read(1,1000) title
      read(1,2000) var,ix,iy,iz,time,xstart,ystart,zstart,dx,dy,dz,
1      (((q(i,j,k),i=1,ix),j=1,iy),k=1,iz)
1000  format(a80)
2000  format(a4,/,3i4,/,4e12.4),/,3e12.4),/,(6e13.6))
```

Although the example above is in FORTRAN, the user can translate the above logic to any desired software capable of reading alphanumeric data. The following table contains the values for the parameter variables in this data set.

The domain grid construction in each coordinate direction relies upon three numbers from the data set header lines: 1) the starting point, 2) the space between each point, and 3) the total number of points for that coordinate direction. Sample logic in FORTRAN for the x direction is as follows:

```
      do i = 1, ix
          x(i) = xstart + (i-1)*dx
      enddo
```

Similarly, replace ix, xstart, and dx with the corresponding variables for the y and z directions to construct the three-dimensional grid.

MATLAB code to read and plot PWS NASA Dataset files

```
% Read in data for one PWS NASA Dataset file.
[InFileName,InPathName] = uigetfile({'*.faa'}, 'Select file');
fidIn = fopen([InPathName InFileName], 'rt');
strTitle = fgetl(fidIn);           % file title
strVarName = fgetl(fidIn);         % variable name
ix = fscanf(fidIn, '%f', 1);       % number of grid points in x
iy = fscanf(fidIn, '%f', 1);       % number of grid points in y
iz = fscanf(fidIn, '%f', 1);       % number of grid points in z
newLineCharacter = fscanf(fidIn, '%c', 1); %ok<*NASGU>
simtime = fscanf(fidIn, '%f', 1);  % time of the simulation in seconds
xstart = fscanf(fidIn, '%f', 1);   % x origin in meters (+x = true east)
ystart = fscanf(fidIn, '%f', 1);   % y origin in meters (+y = true north)
zstart = fscanf(fidIn, '%f', 1);   % z origin in meters (+z = altitude above ground)
newLineCharacter = fscanf(fidIn, '%c', 1);
dx = fscanf(fidIn, '%f', 1);       % grid cell size for x in meters
dy = fscanf(fidIn, '%f', 1);       % grid cell size for y in meters
dz = fscanf(fidIn, '%f', 1);       % grid cell size for z in meters
```

```

newLineCharacter = fscanf(fidIn, '%c', 1);
% Loop through file reading in one line at a time to get the data values.
totValues = ix*iy*iz;
fieldSize = 13;
valuesPerLine = 6;
dataValues = zeros(totValues, 1);
for idx = 1:valuesPerLine:totValues
    % Read in a line of data, add spaces between values, and then read
    % values into vector as floating-point.
    str = fgetl(fidIn);
    k = fix(length(str)/fieldSize);
    str = reshape([reshape(str, fieldSize, k); blanks(k)], 1, length(str)+k);
    dataValues(idx:idx+k-1) = sscanf(str, '%f');
end
fclose(fidIn);

% Plot data for z = 150 meters altitude.
xMeters = (((1:ix)-1) * dx) + xstart;
yMeters = (((1:iy)-1) * dy) + ystart;
zMeters = (((1:iz)-1) * dz) + zstart;
idx = find(zMeters == 150);
dataValues = reshape(dataValues, ix, iy, iz);
figure, pcolor(xMeters, yMeters, dataValues(:, :, idx).'), shading flat
colorbar, xlabel('x (m)'), ylabel('y (m)'), title([strVarName ' - ' strTitle])

```